

Fernando Rojas

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EDUCATION

Universidad Politecnica Taiwan Paraguay | National Taiwan University of Science and Technology

B.S. Information Engineering | 2023-Present

Relevant Coursework: Data Structure and Algorithms, Operating Systems, Artificial Intelligence and Machine Learning, Embedded Systems, Computer Organization and Architecture, Advanced Computer Networks.

SKILLS

Programming Languages: Java, C, C++, C#, Python, SQL, HTML/CSS, Javascript, YAML, Lua.

Frameworks & Tools: Git, Unity, FastAPI, Docker, Wireshark, Flask, Tensorflow, Keras, Docling.

WORK EXPERIENCE

Cast S.A. (Paraguay): *Software Engineering Intern*

December 2024 – February 2025

- Developed an artificial intelligence agent to be implemented as a conversational chatbot to answer client's queries using the open source machine learning RASA framework.
- Conducted an internal survey to gather data about frequently asked questions by clients about the use of the company's software in order to train the agent.

REEKON (Taiwan): *AI Engineering Intern*

March 2026– June 2026

- Refactored the Retrieval-Augmented Generation (RAG) pipeline by integrating Docling for multi-format document ingestion, significantly reducing repository size and achieving a 100% accuracy rate on test-set queries.
- Designed and implemented diverse document chunking strategies alongside automated regression tests to systematically evaluate and optimize future retrieval techniques.
- Tailored targeted retrieval rules specifically governing CTS Mobile Payments to ensure high-precision answers for complex financial/regulatory queries.

International Games System (Taiwan): *Software Engineering Intern*

June 2026 – August 2026

- Optimized and extended the proprietary climate gameplay editor, implementing complex interaction effects between dynamic weather systems and scene elements.
- Architect level flow and boss battle state machines using Unity/game-engine Timelines, successfully delivering the complete logic and flow for the game's third level.
- Refactored legacy codebase utilizing "separation of concerns" principles, drastically reducing script sizes, increasing modularity, and authoring comprehensive English technical documentation.
- Conducted hardware-in-the-loop testing and presented finished systems and level builds directly to the development team.

PROJECTS

An Electroencephalography–based Brain Computer Interface for Real Time Control Using Motor Imagery and Deep Learning [See more](#)

- The project goal is to translate EEG readings into movement commands within a 3D environment by training a model that classifies the “thoughts” of the user. The dataset was built from scratch using scripts to record the electrical activity in the pre-frontal cortex region of the brain.
- Winner of the “Excellent Work” and “Best Popular Project” awards at the Capstone Project Competition 2025.2 for UPTP (Paraguay) students at NTUST (Taiwan).

Reforest AI [See more](#)

- Developed a machine learning pipeline to identify optimal reforestation sites by analyzing satellite imagery (Earth Engine) and the normalized difference vegetation index to highlight biomass density in a region.
- Implemented data processing routines to convert raw satellite imagery into structured 8-channel tensors suitable for the model. The final model achieved 94% in the validation tests.

3D Java Game [See more](#)

- Led the development of a fully single-player 3D blackjack game, integrating game logic with a custom game engine from scratch using Java and LWJGL3 (OpenGL bindings), featuring a complete rendering pipeline with programmable shaders and lights.
- Applied matrix transformations and linear algebra using Java OpenGL Math Library for camera movement, object positioning, and bone animations.

SigmaChat [See more](#)

- Led the development of a real-time chatroom application using Python and Flask, integrating user authentication and a leaderboard for user statistics (using MySQL) and a unique coding competition feature in which players need to provide the correct answer to a given line of C++ code with an error.
- Designed an interactive, low-latency user experience using Socket.IO and MongoDB for chat history and application data, as well as the website appearance using HTML/CSS.

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